BUSINESS DATA MANAGEMENT

Lecture04

Basics of Demand and Supply

Hello, friends. In this particular video, we will try to understand the basic concepts of demand and supply. We'll understand the definition. We will understand what the different forms of demand and supply are. We will understand what the law, if any, is that governs demand and supply.

What is the demand schedule? What is the demand curve? And we'll also try to understand what we mean when we say individual demand versus market demand. So let's get started here. So as you all may have heard me, I mentioned the term market as well in the context of demand and supply.

So what is a market? What do we mean when we say a market? The picture before you will probably give you a good indication of what a marketplace is. A marketplace or a market is where you have a group of buyers and sellers at a given point in time where the buyers are trying to buy products or services and the seller is trying to sell the product or service. So there is this whole dynamic that evolves between buyers and sellers in a particular location or on a particular platform, and that is regarded as a marketplace.

And this marketplace or market has evolved over a period of time as in today's context, the marketplace does not remain only in a physical sense. It moves beyond physical sense into a virtual sense as well. Now that we know what really a market is, it's important to understand that there are different forms of market. For example, there is this whole organized market, which is, as the name suggests, which is organized in terms of there are, it is a homogeneous place where buyers and sellers, they meet at a specific time or place. For example, you have these Mandis,

or you have these farmers market, for example, where there are a group of sellers who are essentially either vendors or farmers who are trying to sell their

commodity. And then you have a certain set of buyers who are there to buy that particular commodity at a given point in time, at a given place, for a particular price. On the other hand, the markets are also highly unorganized. Okay. For example, if you go back in time, you know, as children, you would, you would want, you would see that there would be, you know, different ice cream vendors who are selling ice cream in a particular town or village or a city, and they would be scattered all around, you know, selling different types of ice creams and you would get to choose which product to buy.

You know, it could be at different points in time. It could be a different place. So that's, like a very basic example of what an unorganized market is. More structurally speaking, in today's context, the forms of market can also be defined in terms of the level of competition.

For example, think of a particular product where there are a number of sellers. And of course, there are a number of buyers as well. This kind of a market is regarded as a competitive market where there is a lot of competition between the sellers and there are a number of buyers who are ready to buy from this set of sellers. Here, what is important to understand is that the price of a particular product or a good or a service is determined by the market forces. They are not, I repeat, they are not influenced by a single buyer or a set of buyers.

They will have very little, if at all, impact on the price of that particular good or service. Theoretically speaking, if you look at a perfectly competitive market, there are two distinct characteristics that need to emerge. Number one, The goods offered for sale are the same or similar. There is not too much differentiation between these goods.

They are almost the same. And second is that buyers and sellers are so many that no single individual or a set of individuals can influence the price either on the buyer's side or on the seller's side. I repeat, the first one is that the goods offered for sale or the services offered for sale are largely homogeneous or similar. And second is none of these two entities, buyers or sellers, should be

able to influence the price. No single buyer or seller should have any impact on the final price of the good or service.

Now, that kind of a situation is regarded as a perfectly competitive market. Again, how is the price determined? The price is determined by market forces. And we will discuss all of this in a lot of detail. But this is just to give you a very high level understanding when we say, you know, market for this market demand, market supply.

What does market really mean? Now we talked about this whole perfectly competitive market where there is perfect competition. There are a number of buyers and a number of sellers. However, in reality that may not be the case. You have situations which are where there is just one seller or few sellers.

Now that's not a perfectly competitive market because here the bargaining power of the seller here is extremely high. And such kind of markets where there is only a single player is called as a monopolistic market or a monopoly. We say, hey, in this marketplace, this particular organization has a monopoly. What does that mean?

It means there is a single seller here. Can you think of certain examples? Think about when you need to buy a railway ticket. Where do you go to? There's just one entity there today, which is called the IRCTC.

Not right now, but even a few years back, you had just a few players in the oil and gas industry. For example, Indian Oil Corporation was the sole distributor. That kind of a situation was a monopolistic situation. At the same time, when there are not just one seller, but there are, let's say, a few sellers, not too many, not one, but somewhere in between. Such situations, such kind of market situation are called oligopolistic market or oligopoly.

Here, the market is dominated by a few players. Think in the context of the Indian telecom industry. You do not have hundreds of players. You have less than 10

players who are operating. You can think about your service provider, Vodafone, Idea, Airtel, etc.

So there are not too many players. And in such a situation, the market is regarded as an oligopolistic or an oligopoly market. Why is it important to understand this? It is important to understand because now the pricing is determined not through free market forces. It is influenced by the ones who have higher bargaining power or higher negotiation power.

And when you have fewer players or when you have just one player, the bargaining power is much higher for that particular player or a set of few players. Now that we have understood what market is and what are the different forms of market, it might be very useful to fundamentally understand when somebody says, hey, there is a demand for a particular product or for a particular service, what does that mean? Demand means that the quantity that is ready to be consumed by the buyers They are both willing and able to buy that particular product or a service at different prices during a given period of time. So demand is defined as the quantity of a good or service that the consumers are willing and able to buy at different price points for a given period of time.

So what is important to highlight here are the following. One, demand comes in when there is a willingness to buy, when there is a want or a need to buy that particular good or service. Now that is not enough. Willingness is not enough. Willingness has to be backed by their ability to buy.

That means that they're purchasing power to buy. Different prices. Different prices mean that the demand will change based on the price point. Think about it. If you're trying to buy a particular good or service for a particular price, what happens when the price moves up?

Your demand or your willingness to buy or your even ability to buy will go down. Therefore it is the willingness and ability to buy at different prices for a given period of time because demand is measured over a period of time. Demand is measured for a time period and therefore it becomes important to understand

that demand is something which is the need or the want to buy a particular good or service and having the ability to buy that good or service at different prices for a given period of time. For example, a person may want to buy a BMW or a Mercedes but let's say they are not able to afford it.

Now that is not considered as a demand as per the demand definition in economics. That is just maybe a desire, just a fancy to buy something, a dream to buy something. It is not regarded as demand. So when companies look at their demand analysis, they're looking at customers who are not just willing to buy, but also have the ability to buy the particular good or service at a particular price.

So the key takeaway from this definition, if all of you can make a note of it, is that demand has to be backed by the purchasing power of the consumers. Now, just look at it from an economic standpoint, we have something called the law of demand. The law of demand or the principle of demand states that ceteris paribus, which is other things remaining the same. Okay, other things remain the same when the price of a particular product rises, the quantity demanded for that particular product falls down. For example, if I want to buy a particular mug and the price for this particular mug is, let's say, 50 rupees, maybe I can buy five mugs.

So let's say the price of this particular mug goes up from 50 rupees to 100 rupees. Will I be able to buy the same number of mugs that I was willing to buy earlier? No. Because my ability now to buy goes down. So that is what this means.

This means when the price of a particular good rises, the quantity that is demanded of that particular good falls down. And when the price falls, in that case, other things remaining the same, the quantity that is demanded by the consumer should go up. That's the law of demand. Very simple, very straightforward. When you put this demand in a tabular form, when you put this whole picture in a tabular form in terms of data, that is called the demand schedule.

It's the table that displays this whole relationship between price point, and the quantity of demand. And the price point is on the Y axis and the quantity of demand is on the X axis. And we will talk about that in just a moment. And this representation, this graphical representation of the price of a particular good and the quantity for particular good or service, when you represent it graphically, it is called the demand curve.

And we will just kind of see how this demand curve moves Pictorially and visually. It is important to understand that this particular curve, as we will see in just a moment, is sloping downwards. Let's take a look at that. What is a demand curve?

A demand curve is a graphical representation of the relationship between the price of a particular product and the quantity that is demanded. And this particular curve, which is downward sloping, is called the demand curve. Why is it downward sloping? It is downward sloping because as you can see, the number of goods that are demanded, okay, the number of goods that are demanded, the quantity of the demand goes down. The quantity of the demand goes down as the prices increase.

Okay. Let's take an example. What was, if you look at this particular graph, what is the quantity of demand when the price point of this particular ice cream cone is 1. When the price point of this ice cream cone is 1, the demand here, as you can see, the intersection here between x and y axis is coming to as 8.

8 is the quantity. Now let's say the price point moves from 1 to 3. There is no demand. It becomes 0. Somewhere in between, let's say one and a half, the demand moves from eight, which is at price point one, when the price point moves from one to one and a half, the demand moves down from eight quantities to six quantities. Okay, so this is a graphical representation of this whole data point which illustrates again that the quantity demanded of the good changes as the price changes because the lower price

of a particular good, other things remain the same, will increase the demand for a particular good or service, and when the price point decreases of a particular good or service, the demand of that particular good or service rises. So as you can see, when you look at the demand table, when the price point of an ice cream cone is, let's say, one, the quantity of that particular cone is eight. When the price point moves from 1 to 2, the quantity of the cone demanded moves down from 8 to 4. That is what is called the demand curve. Now, we often talk about demand not in the context of individual.

We don't say demand for this particular individual or that particular individual. We look at demand from the point of view of the total demand or what is also called the market demand. So what is market demand? Market demand is the sum total of the individual demand for a particular product or service. So the quantity that is demanded

When you look at it from a market standpoint, it is a summation of the quantity that is demanded by all the buyers at a given price point for a period of time. Going back to the definition of demand. Now, market demand, like individual demand, will also have a curve. And that curve demonstrates how the total quantity demanded varies based on the price of that particular good or service. Now, the same concept that we looked at from an individual demand standpoint, we will look at it exactly the same from the total demand standpoint.

So here, let's say the quantities for the entire market will be looked at at a given price point. That would be the only difference and not at an individual level. Now, just to put it in perspective, as an example, suppose the price of a particular ice cream cone is two and person A demands four ice cream cones and person B demands three ice cream cones and assume that this is the total market of two individuals for ice cream cone at the price point of two, the total demand that you will be looking at would be the summation of these two individuals' demand, which is four plus three and that is seven. As simple as that.

Now that we've understood conceptually what demand means, it's important to understand the supply side of things. What is supply? Supply is the quantity of the particular good or service that the sellers are willing and they are able to provide or offer for sale at different prices during a given period of time. Again, like the way we deconstructed demand definition, let us try and deconstruct the supply definition. What are the important elements here?

Number one, the seller should be willing to sell. Of course, the seller is in the business. They should have the willingness to sell. But what is more important is they should have the ability to sell. What does that mean?

It means they should have the capacity to sell. For example, if you have a particular manufacturing facility, and you can only produce let's say 10,000 units of let's say some kind of a hardware, some kind of a tool, just 10,000 units. Can you sell beyond 10,000 units? Do you have the ability to sell beyond 10,000 units?

The answer is no. Therefore, to be able to determine supply, you need to figure out what is the ability of the suppliers to be able to sell or offer that particular product or good or service in the market. And this has to be offered at a given price point because your supply quantities will change as we will see in just a moment depending upon different price points. Think about it.

Think about you as a business person. What would you do if the price point of your particular goods or services moves up? If it moves up, that means you are able to make more profits if you are able to sell. That means your willingness will be much higher to be able to offer that particular good or service out in the market as compared to if, let's say for some reason, the price of the particular good or service that you're selling goes down dramatically. Would you have the willingness to sell it?

You would not. Why? Because you might incur a loss at that given low price point. Okay, so higher the prices, higher is the willingness to supply because producers, businesses are motivated by the underlying profit that they will be

able to generate out of this particular sale. Okay, so that's the definition and that's the basic understanding of what we mean when we say supply.

Let's look at supply a little bit more deeply. Let's first try and understand again, what is the law of supply? Like we discussed the law of demand, what is the law of supply? The law of supply is, again, other things remaining the same. Okay, ceteris paribus.

When the price of a particular good or service moves up, the quantity supplied by producers also moves up. Remember when we looked at it from a demand perspective, when the price of a particular good goes up, your willingness and ability to buy that particular good and to demand that particular good goes down. In the supplier's case, it's just the opposite. Here when the price of the good rises, the quantity supplied

the willingness to supply those quantities out in the market also rises. And when the price falls, the quantity supplied by producers will fall as we just discussed, because they will now probably not just make less profit, they might even make a loss. So this happens again because selling becomes more profitable when your price point is higher. That prompts the sellers or that enables or motivates the sellers to produce more and sell more. And mind you, we also discussed when we discussed supply, we said their ability to produce.

Just because the price is moving up, they cannot produce an infinite amount of quantity. They can only produce the amount of quantity for which they would have the capacity to produce. Going back to the example of those 10,000 units that we just discussed. Now that you have this whole relationship of the quantities that are supplied with pricing.

If you put that in a tabular form, that relationship of quantity and price is regarded as the supply schedule. And when you plot that in a graphical form, that is regarded as the supply curve. Again, just like the way we discussed the demand, the market demand, the market supply, is the aggregate of all individual suppliers or sellers together that form the market. So the market supply curve, just like the

way we will discuss the individual supply curve, will show how the quantity supplied responds to changes in the price, assuming all other factors remaining the same.

And let's look at that with the help of this example. Now, the same example that we discussed in the context of demand and we looked at this example of, you know, number of ice cream cones that are demanded and their price points, we saw there was a downward slope. Whereas you look at supply, you will see that the slope is upward. It is the opposite to demand. And why is that?

It is because as and when the prices move upward of this particular product which is the ice cream cone, you will see that the willingness of the sellers or the suppliers to sell those ice cream cones also moves up. So at the price point of one and a half, the willingness and ability to supply was two quantity; at a price point of three, the willingness and the quantities of cone that is supplied moves from two to five. So that's why it's an upward slope. So with that, we will close this particular video.

And in the next video, what we will try and do is we will try and understand the various demand curves and supply curves. We will also bring together these two elements together. OK, so far we've been just looking at the demand independently and supply independently. We will bring both these constructs together and what we will come to the point would be around looking at what is called the equilibrium price. OK, so we'll discuss and understand the concept of equilibrium price as well.

So I look forward to seeing all of you in the next video. Bye for now.